1 電源入力 / POWER INPUT

INPUT:5V_LOGIC,GND

+5V_LOGIC
GND
J_PWR_LOGIC

C1 470uF/10V LOGIC
C2 100nF LOGIC

XH: pin 1 = +5V_LOGIC, pin 2 = GND

JP_STAMP_5V – REMOVE BEFORE USB POWER

+5V_LOGIC
+5V_STAMP

INPUT:5V_SERVO,GND

+5V_SERVO
GND
J_PWR_SERVO

C3 1000uF/10V SERVO

SERVO:5V,GND

+5V_SERVO
GND
J_SERVO_5V_OUT

2 ホスト接続 / SOLIST–AI + STAMP–S3A

SOLIST_CN3_2x7

I2C_SCL
I2C_SDA
TFT_DC
TFT_MOSI
GND
GND
GND
TFT_RST_N
PCA_OE
TFT_SCK
TFT_CS_N

J_SOLIST

CN3 pins 5 and 8: NC

J_STAMP_17

J_STAMP_6

J_STAMP_6 physical pads

18 GND | 20 G0 NC | 22 EN NC

24 G44/WIN_A | 26 G43/BCLK | 28 3V3

MIRRORED CARRIER SOCKETS

Module component side faces up; compare USB and antenna ends before insertion.

3 表示・サーボ / TFT SPI + PCA9685 I2C

TFT:MISO_NC,BL,SCK,MOSI,DC

+5V_LOGIC
TFT_SCK
TFT_MOSI
TFT_DC
J_TFT_SIG

PCA:GND,SCL,SDA,3V3

GND
I2C_SCL
I2C_SDA
+3V3_STAMP
J_SERVO_CTRL

R9 10k TFT_CS pull-up

TFT_CS_N
+3V3_STAMP

R8 10k PCA_OE pull-up

PCA_OE
+3V3_STAMP

TFT:RST,CS,GND,VCC

TFT_RST_N
TFT_CS_N
GND
+5V_LOGIC
J_TFT_PWR

PCA:OE,GND

PCA_OE
GND
J_PCA_OE

TFT is write–only; MISO pin 1 is NC. Verify module 5 V input/BL rating.

OE defaults HIGH: servo outputs disabled until Solist–AI enables them.

4 音声I2S / MICROPHONE + AMPLIFIER

R1 33R BCLK_MIC

I2S_BCLK_SRC
I2S_BCLK_MIC

R3 33R WS_MIC

I2S_WS_SRC
I2S_WS_MIC

AMP:LRC,BCLK,DIN,GAIN

I2S_WS_AMP
I2S_BCLK_AMP
I2S_DOUT_AMP
GND
J_AMP_SIG

R2 33R BCLK_AMP

I2S_BCLK_SRC
I2S_BCLK_AMP

R4 33R WS_AMP

I2S_WS_SRC
I2S_WS_AMP

R5 33R AMP_DIN

I2S_DOUT_SRC
I2S_DOUT_AMP

GND
3V3_STAMP
I2S_DIN_MIC
I2S_BCLK_MIC
I2S_WS_MIC
GND
J_MIC

R7 100k MIC_SD pull-down

AMP signal pin 4 GAIN=GND (12 dB nominal)

I2S_DIN_MIC
GND

J_MIC pin order: 1 GND | 2 3V3 | 3 SD | 4 BCLK | 5 WS | 6 L/R=GND

MIC PIN 2 = 3.3 V ONLY

C4 100nF MIC

+3V3_STAMP
GND

AMP:SD,GND,5V

AMP_SD
GND
+5V_LOGIC
J_AMP_PWR

R6 10k AMP_SD pull-down

AMP_SD
GND

+5V_LOGIC
GND

C5 10uF AMP

5 6点入力 / SIX ACTIVE–LOW INPUTS

WIN_A:1=SIG,2=GND

SW_WIN_A
GND
J_WIN_A

WIN_B:1=SIG,2=GND

SW_WIN_B
GND
J_WIN_B

WIN_C:1=SIG,2=GND

SW_WIN_C
GND
J_WIN_C

DOOR_AB:1=SIG,2=GND

SW_DOOR_AB
GND
J_DOOR_AB

DOOR_BC:1=SIG,2=GND

SW_DOOR_BC
GND
J_DOOR_BC

EXEC:1=SIG,2=GND

EXEC_N
GND
J_EXEC

ALL XH CONNECTORS: pin 1 = signal, pin 2 = GND

Contact closure = LOW. Enable Stamp internal pull–ups and firmware debounce.

6 組立・機能注記 / ASSEMBLY NOTES

PCA9685 channel map: CH0 WIN_A CH1 WIN_B CH2 WIN_C CH3 DOOR_AB CH4 DOOR_BC

M3 H1
M3 H2
M3 H3
M3 H4

Servos connect to the PCA9685 module 3–pin outputs; servo V+ is the separate +5V_SERVO rail.

Startup defaults: AMP_SD LOW (R6); PCA_OE HIGH (R8); TFT_CS_N HIGH (R9).

Before fitting modules: measure MIC 3.3 V, PCA logic 3.3 V, AMP 5 V, servo V+ 5 V; verify XH pin 1.

I2C is shared by Solist–AI, Stamp–S3A and PCA9685. Stamp sends input status/events and optional FFT data.

Functional schematic matched to production PCB netlist
Logic 5 V and servo 5 V are separate rails
Stamp–S3A sockets are mirrored on the carrier PCB
Genuine JST XH; pin order is the harness contract

IchiPing

Sheet: /
File: stamp_s3a_interposer_pcba.kicad_sch

Title: IchiPing Solist–AI / Stamp–S3A Interposer – PCBA

Size: A3	Date: 2026–09–13	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/1